

SIVA DINESH KANAKAMALLA

Embedded Software Engineer | GPON/EPON ONU Firmware | Wi-Fi CAPWAP | Embedded C & Linux

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CORE COMPETENCIES

- GPON/EPON ONU Firmware Development
- Wi-Fi Access Point (CAPWAP) Stack
- 802.11k/v Roaming Technologies
- ITSAR Security Compliance
- Embedded C & Linux Systems
- Binary Reverse Engineering (Ghidra)
- DTLS / Network Security Protocols
- ACL, MAC Filtering, PPPoE, DNS, IGMP
- Device Management & Firmware Upgrade
- Production-Line Provisioning Tools
- Network Management System (NMS)
- Debugging & Reverse Engineering
- Auto Channel Selection & WAN Locking

TECHNICAL SKILLS

Programming

- Embedded C, Python

Protocols

- CAPWAP (RFC 5415/5416), DTLS, 802.11k/v, UART, I2C, SPI, CAN

Embedded Systems

- Embedded Linux, FreeRTOS, POSIX Threads

Networking

- TCP/IP, UDP, HTTP/HTTPS, IGMP, NTP, VLAN, QoS

Tools

- Ghidra, Wireshark, tcpdump, GitHub, Postman, CST Microwave Studio

Hardware

- Wi-Fi AP, XPON ONU Router, Ethernet Switch, STM32G431, BeagleBone White

EDUCATION

PG Diploma – Embedded Systems & Design (PG-DESD)

CDAC, Bangalore

Sep 2023 – Mar 2024 | 81.2%

B.Tech – Electronics & Communication Engineering

RVR & JC College of Engg., Guntur

2019 – 2023 | CGPA: 9.01 / 10

PROFILE SUMMARY

- Senior Development Engineer with hands-on experience in GPON/EPON ONU firmware development, Wi-Fi Access Point (CAPWAP) stack implementation, and networking protocols using Embedded C.
- Skilled in implementing ITSAR-compliant security features including authentication, encryption, logging, password policies, and DoS mitigation for production-grade embedded devices.
- Strong background in embedded systems and Linux with proven expertise debugging complex issues related to ACL, MAC filtering, PPPoE, DNS, and IGMP across real-world deployments.
- Experienced in reverse engineering binary/executable files using Ghidra and reconstructing original code logic with AI agents; strong analytical and problem-solving skills delivering optimized, security-certified embedded solutions.

WORK EXPERIENCE

Smartlink Holdings Ltd (Client: Digisol Systems Ltd)

Senior Development Engineer | Bangalore | April 2024 – Present

Promoted from Development Engineer to Senior Development Engineer in June 2026

Wi-Fi Access Point (AP) — CAPWAP Development

- Analysed CAPWAP RFC 5415 & RFC 5416 and developed the full CAPWAP stack for the AP (WTP), enabling controller-managed wireless operations.
- Implemented configuration and device management features including Reboot, Restart, Factory Reset, and Firmware Upgrade via CAPWAP control messages.
- Decoded and debugged vendor-specific payload elements to achieve interoperability between the AP and an open-source CAPWAP AC server.
- Implemented DTLS security for the WTP and deployed the AC server with DTLS support to meet ITSAR security certification requirements.
- Worked on 802.11k/v roaming technologies for the Wi-Fi AP to improve client mobility and seamless handoff performance.

XPON ONU Firmware Development

- Developed and enhanced embedded C firmware for XPON ONU with a focus on security, stability, and performance.
- Integrated secure NTP client (Chrony) with authentication and vulnerability safeguards.
- Implemented vendor-specific EPON/OLT configuration modules for WAN, Wi-Fi (2.4G/5G), and TR-069 control.
- Designed Auto Channel Selection logic for 2.4G and 5GHz bands; implemented WAN IP Locking for Router and PPPoE modes.
- Built MAC Programming utility for production-line provisioning; ensured full production validation including Wi-Fi calibration, throughput, and MAC verification.

ITSAR Security Compliance for ONU

- Applied comprehensive password policy controls, first-login reset, expiration, history validation, and automatic expiry alerts.
- Enforced secure privilege escalation rules, session timeout handling, and restricted file access for the web interface.
- Added MAC filtering, DoS-resistant packet rate limiting (ICMP/UDP/FIN/SYN), AES-128 Wi-Fi encryption, and WPS brute-force protection.
- Debugged and resolved critical stability issues related to ACL, DNS, PPPoE flows, MAC filtering, and internet reachability logic.

Network Management System (NMS) for Ethernet Switches

Apr 2024 – Sep 2024

- Developed Python-based intermediary Switch Server using HTTP/HTTPS for centralized device management.
- Implemented features: Restart, Factory Reset, Firmware Upgrade, Backup/Restore, and Log Retrieval.
- Designed configurations for IP, VLAN, QoS, Port Settings, Isolation, and Mirroring; enabled device discovery, digest authentication, keep-alive, and server-side log rotation.

✔ STRENGTHS

- Strong analytical and debugging capabilities
- Practical embedded systems and networking knowledge
- Effective team player and independent contributor
- Clear documentation and reverse-engineering experience

Ⓐ LANGUAGES

- English, Telugu

📄 PERSONAL DETAILS

Location: Bangalore, India
Notice Period: 30 Days

📁 HIGHLIGHT PROJECTS & INNOVATIONS

CAPWAP Stack & ITSAR-Compliant Wi-Fi AP

Smartlink / Digisol Systems

- End-to-end CAPWAP (RFC 5415/5416) implementation for controller-managed Wi-Fi Access Points with DTLS security and 802.11k/v roaming, enabling full ITSAR certification readiness.

XPON ONU Firmware & Security Hardening

Smartlink / Digisol Systems

- Production-grade embedded C firmware for XPON ONU covering WAN, Wi-Fi, TR-069, secure NTP, Auto Channel Selection, WAN IP Locking, and comprehensive ITSAR security controls (password policy, MAC filtering, DoS rate limiting, AES-128).

Binary Reverse Engineering with Ghidra & AI Agents

Smartlink / Digisol Systems

- Reverse engineered binary/executable files using Ghidra and reconstructed the original code logic with the assistance of AI agents.

CAN-Centric Industrial Monitoring and Control

CDAC | Dec 2023 – Feb 2024

- Built a two-node industrial monitoring system using STM32G431 (ARM Cortex-M4) boards communicating via FDCAN.
- Collected data from MQ2 gas, BMP180 temperature/pressure, and LDR sensors on the TX unit and transmitted over CAN.
- Implemented RX-side control actions: Fan activation on high temperature, Buzzer alert on gas detection, and LED control based on light intensity.
- Displayed real-time sensor values on LCD; integrated ADC, I2C, and GPIO interfaces for sensors and actuators with threshold-based automation.
- GitHub: <https://github.com/sivadinesh2001>

★ AWARDS & RECOGNITION

Best Innovation of the Year 2025–26

- Recognised for outstanding contributions to ONU Firmware Development at Smartlink Holdings Ltd.